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Module 3 Scientific Writing

Reading & Synthesizing papers

24 September 2026 - Fonso Pica



Outline

Part 1: Efficient reading & Paper structure

Common tools for reference management and exploring literature

Part 2: Synthesizing literature using a synthesis matrix

Identifying research gaps

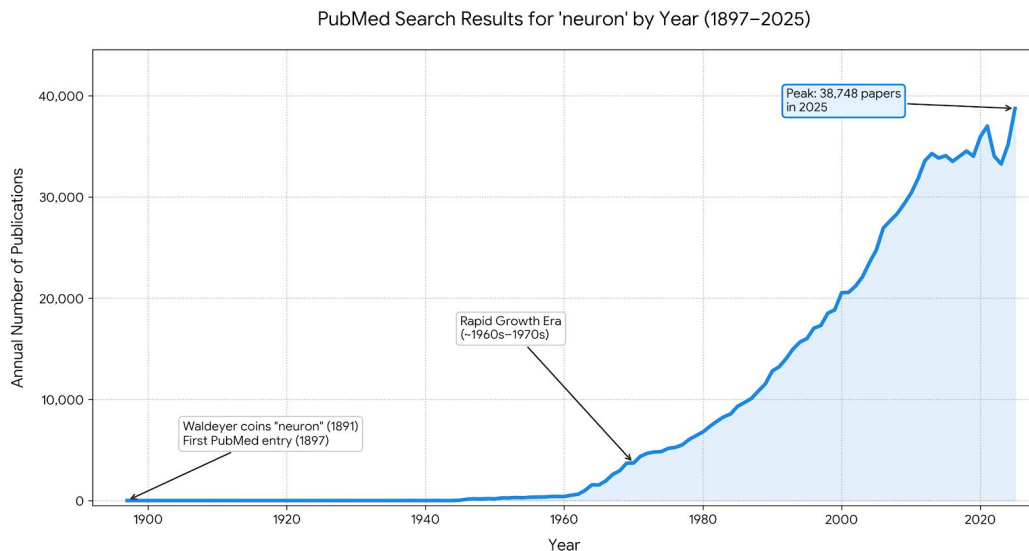
Challenges with reading “the literature”

Keeping up with the field, learning about new techniques, findings, questions, etc.

Remembering what we've read

Understanding what we're reading

Understanding the field



[More about Waldeyer-Hartz](#)

Data from [pubmed](#)



Efficient reading: know your goal

Query-driven approach: reading strategy depends on your goal.

Are you working on an experiment / writing a paper / giving a lecture / ...?

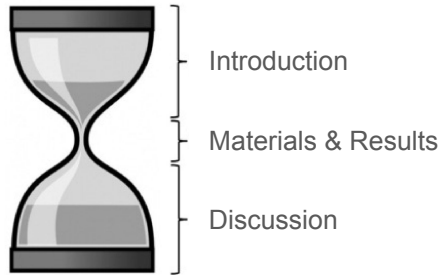
Do you want to learn more about a method / theory / state-of-the art / ...?

Will the outcome be a summary / implementation / inspiration / ...?

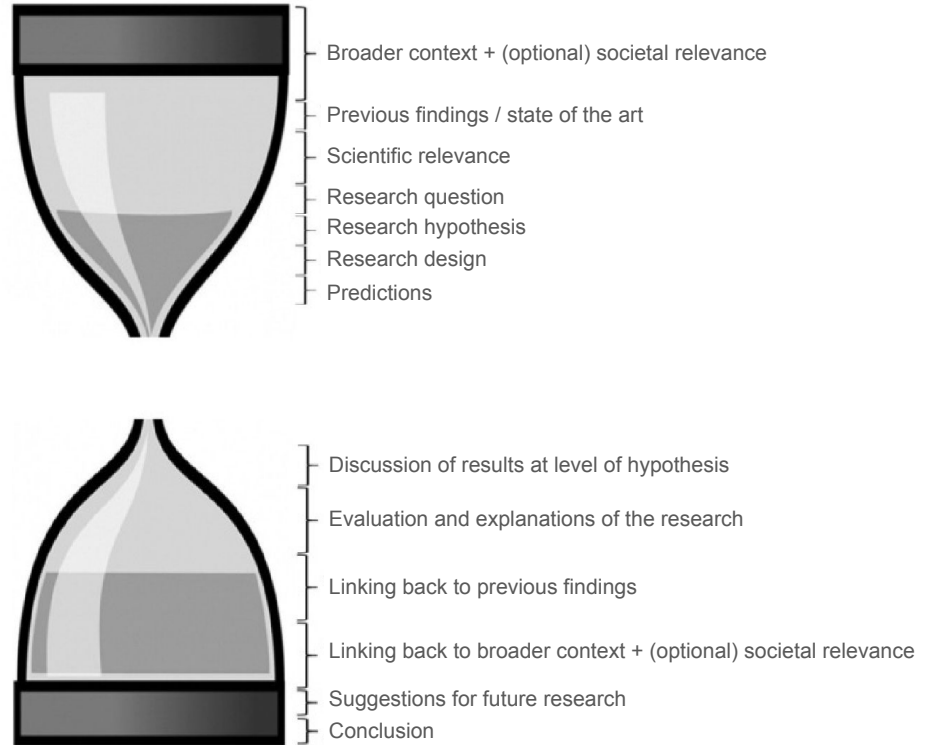
You know what you want, you have a paper, how do you go through it efficiently?

Efficient reading: paper structure

Know where to look



*width represents scope





Efficient reading: Relevant? Skim. Discussion.

Step 1: Is the reference relevant? Read the title and abstract: look for the research question and conclusion then decide [reject / continue] (~5 mins)

Step 2: Skimming - look at the figures and tables, how do the findings relate to the research question? Note your questions and conclusions. (~20 mins)

Step 3: Review the discussion: summary of results and conclusions can often be found at the start. Look for links to previous studies / broader literature. What limitations do the authors address? (~10 mins)

Remember: reading strategy depends on your goal.

Efficient reading: abstract structure

Annotated example taken from *Nature* 435, 114–118 (5 May 2005).

One or two sentences providing a **basic introduction** to the field, comprehensible to a scientist in any discipline.

Two to three sentences of **more detailed background**, comprehensible to scientists in related disciplines.

One sentence clearly stating the **general problem** being addressed by this particular study.

One sentence summarizing the main result (with the words “**here we show**” or their equivalent).

Two or three sentences explaining what the **main result** reveals in direct comparison to what was thought to be the case previously, or how the main result adds to previous knowledge.

One or two sentences to put the results into a more **general context**.

Two or three sentences to provide a **broader perspective**, readily comprehensible to a scientist in any discipline, may be included in the first paragraph if the editor considers that the accessibility of the paper is significantly enhanced by their inclusion. Under these circumstances, the length of the paragraph can be up to 300 words. (This example is 190 words without the final section, and 250 words with it).

During cell division, mitotic spindles are assembled by microtubule-based motor proteins^{1,2}. The bipolar organization of spindles is essential for proper segregation of chromosomes, and requires plus-end-directed homotetrameric motor proteins of the widely conserved kinesin-5 (BimC) family³. Hypotheses for bipolar spindle formation include the ‘push–pull mitotic muscle’ model, in which kinesin-5 and opposing motor proteins act between overlapping microtubules^{2,4,5}. However, the precise roles of kinesin-5 during this process are unknown. Here we show that the vertebrate kinesin-5 Eg5 drives the sliding of microtubules depending on their relative orientation. We found in controlled *in vitro* assays that Eg5 has the remarkable capability of simultaneously moving at $\sim 20 \text{ nm s}^{-1}$ towards the plus-ends of each of the two microtubules it crosslinks. For anti-parallel microtubules, this results in relative sliding at $\sim 40 \text{ nm s}^{-1}$, comparable to spindle pole separation rates *in vivo*⁶. Furthermore, we found that Eg5 can tether microtubule plus-ends, suggesting an additional microtubule-binding mode for Eg5. Our results demonstrate how members of the kinesin-5 family are likely to function in mitosis, pushing apart interpolar microtubules as well as recruiting microtubules into bundles that are subsequently polarized by relative sliding. We anticipate our assay to be a starting point for more sophisticated *in vitro* models of mitotic spindles. For example, the individual and combined action of multiple mitotic motors could be tested, including minus-end-directed motors opposing Eg5 motility. Furthermore, Eg5 inhibition is a major target of anti-cancer drug development, and a well-defined and quantitative assay for motor function will be relevant for such developments.

Source: <https://www.nature.com/documents/nature-summary-paragraph.pdf>



Efficient reading: deeper reading

Review the results: do the conclusions match the results and what are the underlying assumptions that the authors' relied on?

Review the methods: are you familiar with the methods? Are they novel or standard in the field? Is it clear why the methods were used?

Review the limitations: look whether authors address limitations in discussion. Do they offer suggestions for improvement / future research?

Will the outcome be a summary / implementation / inspiration / ...?



Common reference managers

Zotero (zotero.org) - free, with paid cloud storage

Mendeley (mendeley.com) - paid

EndNote (endnote.com) - paid

PaperPile (paperpile.com) - paid

Closer look: Zotero

Targeting light-gated chloride channels to neuronal somatodendritic domain reduces their excitatory effect in the axon

Title	Creator	Year	Date Added
Targeting light-gated chloride channels to neuronal somatodendritic domain reduces their excitatory effect in the axon	Messier et al.	2018	17/04/2022...
Chloride conducting light activated channel ...	Malyshev et al.	2017	17/04/2022...
High-efficiency optogenetic silencing with s...	Mahn et al.	2018	15/04/2022...
Inferring single-trial neural population dyna...	Pandarinath et al.	2018	02/04/2022...
Stochastic activity in low-rank recurrent neu...	Mastrogiuseppe ...	2025	02/04/2022...
Synaptic E-I Balance Underlies Efficient Neu...	Zhou and Yu	2018	01/04/2022...
Distinct Spatiotemporal Response Propertie...	Maor et al.	2016	01/04/2022...
Rapid rebalancing of co-tuned ensemble ac...	Kang et al.	2025	01/04/2022...
Principles of cortical interactions in modular re...	Kong et al.	2026	01/04/2022...
Feature-specific inhibitory connectivity augme...	Ogando et al.	2025	01/04/2022...
Neural oscillations as a signature of efficient c...	Chalk et al.	2016	31/03/2022...
Biologically informed cortical models predict o...	Sourmpis et al.	2026	31/03/2022...
Rhythmic circuit function is more robust to cha...	Fournier et al.	2025	31/03/2022...
Annual Incidence of First Silent Stroke in the U...	Leary and Saver	2003	31/03/2022...
Tuned thalamic excitation is amplified by visual...	Lien and Scanziani	2013	31/03/2022...
A High-Light Sensitivity Optical Neural Silence...	Han et al.	2011	31/03/2022...
Flow of Cortical Activity Underlying a Tactile D...	Guo et al.	2014	31/03/2022...
Synaptic mechanisms of top-down control in t...	Oberle et al.	2022	31/03/2022...
Stimulus-dependent differences in cortical ver...	Cone et al.	2024	31/03/2022...
Neural coding during active somatosensation r...	O'Connor et al.	2013	31/03/2022...
Rapid compensatory plasticity revealed by dyn...	Andrei et al.	2023	31/03/2022...
Heterogeneous side effects of cortical inactiva...	Andrei et al.	2021	31/03/2022...
Spatiotemporal constraints on optogenetic ina...	Li et al.	2019	31/03/2022...
Chaos in Neuronal Networks with Balanced Ex...	Van Vreeswijk an...	1996	30/03/2022...

Info

Item Type: Journal Article

Title: Targeting light-gated chloride channels to neuronal somatodendritic domain reduces their excitatory effect in the axon

Author: Messier, Jessica E

Author: Chen, Hongmei

Author: Cai, Zhao-Lin

Author: Xue, Mingshan

Publication: eLife

Publisher: Place

Date: 2018

Volume: 7

Issue: Section

Part Number: Part Title

Pages: e38506

Series: Series Title

Series Text: Journal Abbr: eLife

DOI: 10.7554/eLife.38506

Citation Key: URL: https://pmc.ncbi.nlm.nih.gov/articles/

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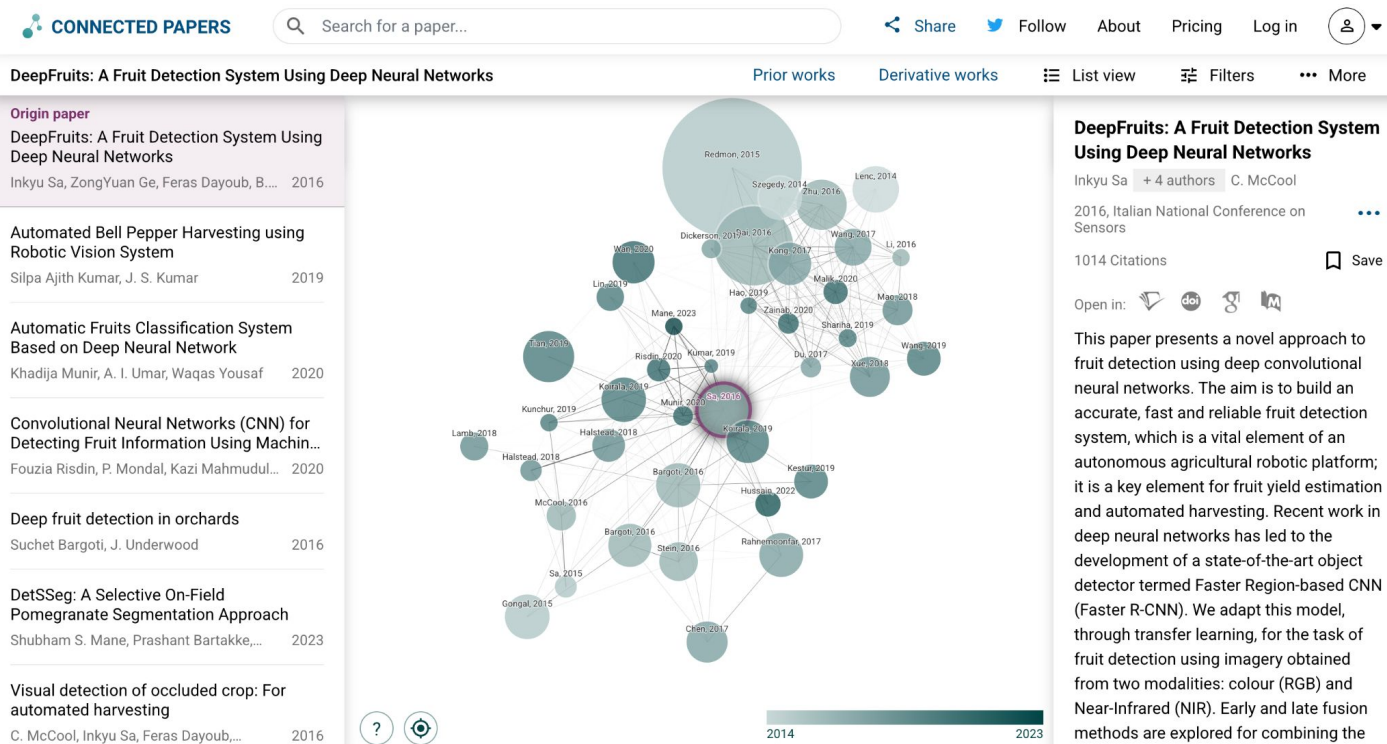
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End of part 1



From reading papers to synthesizing literature

A single paper can only cover so much → build deeper understanding of subject by considering papers together (synthesizing, finding structure in the literature)

Some key questions:

What is known, what is not known, where is more research needed?

Which things are broadly agreed upon (consensus)?

Which things are disagreed upon? (potential research gaps)

Which theories keep coming up? (how are theories connected?)

Which methods have been used? (lack of methodological diversity is itself a gap)

A synthesis matrix

Tool for making a structured literature database.

Can help identify *consensus*, *contradictions*, *trade-offs* and *gaps*.

Can serve as your point of reference during writing.

Subject					
Paper (Author, Year)	Methods	Research question / Hypothesis	Key findings	Limitations	Line of research
Paper 1					
Paper 2					
...					

Synthesis matrix: some common comparison axes

Column Header	Description	Example
Reference (Author, Year)	<i>Who wrote it and when?</i>	Zhang et al. (2021)
Research question / Hypothesis	<i>What specific idea or mechanism were they testing?</i>	PFC parvalbumin (PV) interneurons drive visual gain control during spatial attention.
Methods	<i>What species, tool, and task did they use?</i>	Mouse (<i>in vivo</i> 2-photon imaging + optogenetics during visual discrimination task).
Key findings (Data)	<i>What was the main empirical outcome?</i>	Optogenetic stimulation of PFC PV cells enhanced V1 neural responses to target stimuli by 35%.
Limitations	<i>What did the study fail to answer or control for?</i>	Tested static visual stimuli only; dynamic environments were not evaluated.
Line of Research	<i>What sub-topic or theme does this belong to?</i>	Circuit Mechanisms (Rodent)



Brief: contradictions and research gaps

Contradictions: Two references have conflicting *Key Findings*. Can the *Methods* explain the split?

Scale difference: One set of references covers cellular mechanisms (e.g., rodent optogenetics), while another covers system-level phenomena (e.g., human fMRI).

Common limitations: Do the references share *Limitations*? For instance artificial setups, acute timelines, single species, or single sexes.

Gap: The unresolved mechanism explaining why the results diverge.

Gap: The missing bridge linking the micro-circuit mechanism to the macro-level observation.

Gap: Testing whether the established findings hold true under different conditions e.g. dynamic, real-world, or chronic.

See slide 22 for more examples.

Synthesis matrix - Example 1

Consensus

PFC to visual modulation exists across rodents, primates and humans.

Methodological trade-offs

Human studies provide network-level description but lack causal/cellular detail, while rodent studies lack complex stimuli.

Gap (scale difference)

No study established whether cell-type specific PFC pathways mediate the synchrony observed in humans during complex tasks.

The role of prefrontal cortex (PFC) top-down control in visual attention

Paper (Author, Year)	Methods	Research question / Hypothesis	Key findings	Limitations	Line of research
Zhang et al. (2021)	Mouse (in vivo 2-photon imaging)	PFC PV interneurons drive visual gain control during spatial attention.	Optogenetic stimulation of PFC PV cells enhanced V1 responses to target stimuli.	Used static visual stimuli only; unclear if findings generalize to natural conditions.	Circuit mechanisms (rodent)
Kovacs et al. (2023)	Human (fMRI + EEG)	Frontal-visuocortical network synchrony predicts performance in feature-based attention.	High alpha-band phase synchrony between dlPFC and visual cortex predicted target detection accuracy.	Correlation only; cannot determine causality or underlying cell-type specificity.	Macro-scale networks (Human)
Chen & Liu (2024)	Macaque (Multi-unit laminar recordings)	FEF feedback selectively modulates deep layers of area V4.	Microstimulation of FEF led to layer-specific modulation in V4 during spatial attention tasks.	Small sample size (n=2). Did not record from intermediate cortical areas.	Circuit mechanisms (Primate)



Suggested exercise

Complete a synthesis matrix for 3-5 references

Try to identify 2 research gaps (see next 2 slides)

Subject						
Paper (Author, Year)	Methods	Research question / Hypothesis	Key findings	Limitations	Line of research	...
Paper 1						
...						



Step-by-step: identify research gaps using a synthesis matrix

1. Fill out the synthesis matrix.

2. Do vertical and horizontal scans:

Vertical scan:

- Scan down *Limitations* for recurring unaddressed variables.
- Scan down *Methods* for over-reliance on techniques and model systems.

Horizontal scan:

- Scan across rows to understand how a study's conclusion was shaped by the assumptions or methodology.

3. Look for indicators of gaps:

- **Contradictions:** Two rows show conflicting results in the *Key Findings* column. Look at their *Methods* column to see what difference may cause the split.
- **Scale difference:** One set of rows covers cellular mechanisms (e.g., rodent optogenetics), while another covers system-level phenomena (e.g., human fMRI).
- **Common limitations:** Scanning down the *Limitations* column reveals that every study relies on static paradigms, acute timelines, single species, or single sexes.



Extended list of common research gaps

Contradiction: Divergent results caused by subtle methodological variations.

Scale differences: Missing bridge between micro-level (cellular) and macro-level (systems/behavior) findings.

Methodological boundary: Unexamined assumptions due to universal reliance on a single paradigm.

Innovation lag: Outdated paradigms that have not been tested with modern high-resolution tools.

Ecological Validity: Mechanisms demonstrated in artificial setups that may not hold in naturalistic conditions.

Population / Demographic: Oversimplified generalizations based on narrow sample demographics (e.g., single-sex or single-age cohorts).

Structure-Function relation: Anatomically or structurally mapped pathways lacking physiological or functional validation.



Summary

Efficient reading

- Query-driven: relevant? [Yes: read / No: skip]
- Skim: 1. Abstract, 2. Figures, 3. Discussion
- Remember: reading strategy depends on goal

Synthesis matrix

- Structured reference database
- Helps to identify consensus and gaps

Synthesis matrix - Example 2

Evolution of paradigm

The field shifted from viewing dopamine as a uniform scalar error signal (Schultz) to a rich, high-dimensional probability distribution (Dabney) and a dynamic motivator for action (Hamid).

Methodological trade-offs

Classic single-unit recordings provided detailed temporal firing dynamics (Schultz), but modern optics/neuropixels (Dabney, Hamid) are required to dissect cell-type heterogeneity and population-level spatial distributions.

Gap

It remains unknown how distributional dopamine signals (Dabney) are integrated at the level of downstream medium spiny neurons (MSNs) in the striatal circuit to update tonic value representations and regulate immediate action selection (Hamid).

The role of dopamine in reinforcement learning					
Paper (Author, Year)	Methods	Research question / Hypothesis	Key findings	Limitations	Line of research
Schultz et al. (1997)	Non-human primates (<i>in vivo</i> single-unit electrophysiology)	Midbrain dopamine (DA) neurons encode temporal difference (TD) Reward Prediction Error	DA neurons phasically fire to unexpected rewards, shift to predictive cues after learning, and dip below baseline when expected rewards are omitted.	Cannot isolate cell-type specificity; focuses purely on scalar (single value) mean predictions rather than full probability distributions.	Classical RPE Hypothesis (Scalar)
Hamid et al. (2016)	Mouse (<i>in vivo</i> voltammetry / fiber photometry in striatum)	Striatal DA signals rapidly track value dynamically to motivate action, separate from cue-driven phasic RPEs.	Fluctuations in striatal DA concentrations (tonic/slow waves) dynamically scaled with the local value of the environment and correlated with decision speed/effort.	Fast-scan cyclic voltammetry lacks single-cell resolution; difficult to untangle whether tonic DA reflects value tracking or dynamic RPE over short timescales.	Motivation vs. Learning (Tonic DA)
Dabney et al. (2020)	Mouse (<i>in vivo</i> optogenetics + high-density Neuropixels)	The brain uses <i>distributional</i> reinforcement learning, where individual DA neurons code different optimistic / pessimistic RPEs.	Different DA neurons showed distinct reversal points and response amplitudes to the same reward probability, representing a full distribution of outcomes.	Task relied on passive conditioning; unclear how distributional DA signals modulate downstream striatal target neurons during active, complex decision-making.	Distributional RL Hypothesis